

COUNTEREXAMPLES TO AN ALL-POSITIVE BRANNAN COEFFICIENT INEQUALITY

ABSTRACT. For positive parameters α, β , let $A_n(x; \alpha, \beta)$ be defined by $(1 + xz)^\alpha(1 - z)^{-\beta} = \sum_{n \geq 0} A_n(x; \alpha, \beta)z^n$. We give an exact counterexample to the assertion, recorded in Hayman–Lingham Problem 5.44, that $|A_j(x; \alpha, \beta)| \leq A_j(1; \alpha, \beta)$ for every odd $j \geq 3$, every $|x| = 1$, and all $\alpha, \beta > 0$. At $(\alpha, \beta, j, x) = (3/2, 1/14, 3, -1)$, the two sides are $20/343$ and $33/686$, with reverse gap $1/98$. We also classify the endpoint inequality completely when $j = 3$ and construct positive-right-hand-side counterexamples at every integer index $j \geq 3$.

1. THE RECORDED PROBLEM AND ITS SCOPE

For $\alpha, \beta > 0$, define the coefficient polynomials by the analytic binomial series at the origin,

$$(1) \quad (1 + xz)^\alpha(1 - z)^{-\beta} = \sum_{n \geq 0} A_n(x; \alpha, \beta)z^n.$$

Hayman and Lingham record the following all-positive generalization in Problem 5.44 of [4]: does

$$(2) \quad |A_j(x; \alpha, \beta)| \leq A_j(1; \alpha, \beta)$$

hold for every $\alpha, \beta > 0$, every odd $j \geq 3$, and every $|x| = 1$? Their text emphasizes that even the generalized $j = 3$ case was unknown. This formulation is broader than the standard modern Brannan coefficient conjecture, which restricts the parameters to the unit square; see, for example, [5, 1, 2, 3]. The latter parameter range is not challenged here.

Our first result answers the literal all-positive question negatively.

Theorem 1.1 (Exact counterexample). *The universal assertion (2) is false. More precisely,*

$$(\alpha, \beta, j, x) = \left(\frac{3}{2}, \frac{1}{14}, 3, -1\right)$$

satisfies all its hypotheses, whereas

$$A_3\left(1; \frac{3}{2}, \frac{1}{14}\right) = \frac{33}{686} > 0, \quad \left|A_3\left(-1; \frac{3}{2}, \frac{1}{14}\right)\right| = \frac{20}{343} = \frac{40}{686}.$$

The reverse gap is exactly $1/98$.

The witness has $\alpha = 3/2 > 1$. It therefore lies outside the unit square on which the classical conjecture is known, while remaining inside the unrestricted positive quadrant of the recorded question.

2. DEGREE THREE

Finite coefficient extraction from (1) gives

$$(3) \quad A_3(x) = \frac{\beta(\beta + 1)(\beta + 2)}{6} + \frac{\alpha\beta(\beta + 1)}{2}x + \frac{\alpha(\alpha - 1)\beta}{2}x^2 + \frac{\alpha(\alpha - 1)(\alpha - 2)}{6}x^3.$$

We suppress the parameters when no ambiguity can arise.

Lemma 2.1. *Put $s = \alpha + \beta$ and $d = \alpha - \beta$. Then*

$$A_3(1; \alpha, \beta) = \frac{s(s^2 - 3d + 2)}{6}, \quad A_3(-1; \alpha, \beta) = -\frac{d(d - 1)(d - 2)}{6}.$$

Proof. The first identity follows by summing (3) and collecting in s, d . For the second, setting $x = -1$ in (1) yields the identity of normalized analytic germs

$$(1 - z)^\alpha(1 - z)^{-\beta} = (1 - z)^d.$$

The coefficient of z^3 is therefore $-\binom{d}{3}$. □

Proof of Theorem 1.1. At the stated parameters, $s = 11/7$ and $d = 10/7$. Lemma 2.1 gives

$$A_3(1) = \frac{11}{42} \left(\frac{121}{49} - \frac{30}{7} + 2 \right) = \frac{33}{686}, \quad A_3(-1) = -\frac{(10/7)(3/7)(-4/7)}{6} = \frac{20}{343}.$$

Consequently $|A_3(-1)| - A_3(1) = (40 - 33)/686 = 1/98 > 0$, as claimed. $\square$

The endpoint phenomenon admits a complete two-parameter classification.

Theorem 2.2 (Endpoint classification). *For every $\alpha, \beta > 0$,*

$$|A_3(-1; \alpha, \beta)| \leq A_3(1; \alpha, \beta)$$

if and only if

$$(4) \quad F(\alpha, \beta) := 3\beta(\beta + 1) + (\alpha - 1)(\alpha - 2) \geq 0.$$

Thus strict endpoint failure occurs exactly when

$$1 < \alpha < 2, \quad 3\beta(\beta + 1) < (\alpha - 1)(2 - \alpha).$$

Equality occurs exactly on $F = 0$, and throughout the strict failure region

$$(5) \quad |A_3(-1)| - A_3(1) = -\frac{\alpha}{3}F(\alpha, \beta).$$

Proof. Write $A_3(x) = c_0 + c_1x + c_2x^2 + c_3x^3$, with the coefficients from (3), and set $E = c_0 + c_2$, $O = c_1 + c_3$. Direct factorization gives

$$E = \frac{\beta}{6}(3\alpha^2 - 3\alpha + \beta^2 + 3\beta + 2), \quad O = \frac{\alpha}{6}F(\alpha, \beta).$$

The first bracket equals

$$3 \left(\alpha - \frac{1}{2} \right)^2 + \frac{5}{4} + \beta^2 + 3\beta,$$

so $E > 0$. Since $A_3(1) = E + O$ and $A_3(-1) = E - O$, one has

$$|E - O| \leq E + O \iff O \geq 0.$$

Indeed, the implication to the right is immediate when $O < 0$, because then $|E - O| - (E + O) = -2O > 0$; the converse is the triangle inequality for the two nonnegative numbers E, O . This proves the equivalence and shows that equality is exactly $O = 0$. When $O < 0$, the same calculation gives $|A_3(-1)| - A_3(1) = -2O = -\alpha F/3$. Finally, $F < 0$ is possible precisely for $1 < \alpha < 2$, where it is equivalent to the displayed strict inequality. $\square$

For completeness, the portion of the endpoint failure region having a positive right-hand side can also be written exactly. When $1 < \alpha < 2$, define

$$b_R(\alpha) = \frac{\sqrt{24\alpha + 1} - (2\alpha + 3)}{2}, \quad b_F(\alpha) = \frac{\sqrt{1 + \frac{4}{3}(\alpha - 1)(2 - \alpha)} - 1}{2}.$$

Then $0 < b_R(\alpha) < b_F(\alpha)$, and

$$A_3(1) > 0 \quad \text{and} \quad |A_3(-1)| > A_3(1) \iff b_R(\alpha) < \beta < b_F(\alpha).$$

This follows by viewing the bracket in

$$A_3(1) = \frac{\alpha + \beta}{6}(\beta^2 + (2\alpha + 3)\beta - (\alpha - 1)(2 - \alpha))$$

as a strictly increasing function of $\beta > 0$, and observing that on the wall $F = 0$ one has $A_3(1) = E > 0$.

3. POSITIVE-RIGHT-HAND-SIDE FAILURES AT EVERY INDEX

The counterexample is not an isolated degree-three accident.

Theorem 3.1. *For every integer $j \geq 3$, there exist rational $\alpha, \beta > 0$ such that*

$$A_j(1; \alpha, \beta) > 0 \quad \text{and} \quad |A_j(-1; \alpha, \beta)| > A_j(1; \alpha, \beta).$$

In particular, every odd index allowed by Problem 5.44 admits an open set of counterexamples.

Proof. Put

$$m = j - 2, \quad C = m2^{m+1} + 1, \quad c = \frac{3}{2C},$$

and, for $\varepsilon > 0$, set $\alpha = m + \varepsilon$ and $\beta = c\varepsilon$. Let $R_j(\varepsilon) = A_j(1; m + \varepsilon, c\varepsilon)$. The coefficient extraction is finite, so R_j is a polynomial in ε , with $R_j(0) = 0$.

Using

$$A_j(1; \alpha, \beta) = \sum_{k=0}^j \binom{\alpha}{k} \frac{(\beta)_{j-k}}{(j-k)!},$$

where $(\beta)_r$ is the rising factorial, differentiation at $\varepsilon = 0$ gives

$$(6) \quad R'_j(0) = -\frac{1}{j(j-1)} + c \sum_{k=0}^m \frac{\binom{m}{k}}{j-k}.$$

Here the first term is the derivative at m of $\binom{\alpha}{m+2}$, while $(c\varepsilon)_r/r! = c\varepsilon/r + O(\varepsilon^2)$ for $r \geq 1$. After substituting $r = m - k$,

$$\sum_{k=0}^m \frac{\binom{m}{k}}{m+2-k} = \sum_{r=0}^m \frac{\binom{m}{r}}{r+2} = \int_0^1 x(1+x)^m dx = \frac{m2^{m+1} + 1}{(m+1)(m+2)} = \frac{C}{j(j-1)}.$$

Thus (6) becomes

$$R'_j(0) = \frac{cC - 1}{j(j-1)} = \frac{1}{2j(j-1)} > 0,$$

and hence

$$(7) \quad A_j(1) = \frac{\varepsilon}{2j(j-1)} + O(\varepsilon^2).$$

At $x = -1$, the analytic-germ identity gives

$$A_j(-1) = (-1)^j \binom{m + (1-c)\varepsilon}{j}.$$

For sufficiently small positive ε , the upper argument lies strictly between m and $m+1$; consequently this binomial coefficient has exactly one negative factor. Therefore

$$(8) \quad |A_j(-1)| = -\binom{m + (1-c)\varepsilon}{j} = \frac{1-c}{j(j-1)}\varepsilon + O(\varepsilon^2).$$

Subtracting (7) from (8) yields

$$|A_j(-1)| - A_j(1) = \frac{C-3}{2Cj(j-1)}\varepsilon + O(\varepsilon^2) > 0$$

for all sufficiently small $\varepsilon > 0$, because $C \geq 5$. At the same time (7) makes $A_j(1) > 0$. Choosing a sufficiently small rational ε gives rational α, β . Strictness and continuity then give an open neighborhood of counterexamples. $\square$

4. CONTEXT AND LIMITATIONS

The proof of Theorem 1.1 is elementary and symbolic; no bounded computation is load-bearing. Exact arithmetic software may independently replay the displayed witness, but such a replay is corroboration rather than a substitute for the proof.

A documented search through August 6, 2026 found no earlier publicly retrievable counterexample identical to or stronger than Theorem 1.1, nor the endpoint classification or every-index construction above. The results therefore appear new with moderate confidence. Absolute historical priority is not claimed: complete older theses and dissertations of U. C. Jayatilake, related seminar materials, restricted older chapters and reports, and private or non-digitized sources could not all be retrieved. The coefficient identities, endpoint decomposition, and perturbative analysis use known tools; the apparent novelty lies in the negative result and the exact classifications, not in a new general algebraic method.

Finally, the conclusion should not be misread as a counterexample to the standard unit-square Brannan conjecture. The witness has $\alpha > 1$, and the present result answers only the broader all-positive question recorded by Hayman and Lingham.

REFERENCES

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